

OpenStax Calculus Volume 2

Section 3.2 Exercises: Trigonometric Integrals

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Fill in the blank to make a true statement.

1. $\sin^2 x + \underline{\hspace{2cm}} = 1$

2. $\sec^2 x - 1 = \underline{\hspace{2cm}}$

Use an identity to reduce the power of the trigonometric function to a trigonometric function raised to the first power.

3. $\sin^2 x = \underline{\hspace{2cm}}$

4. $\cos^2 x = \underline{\hspace{2cm}}$

Evaluate each of the following integrals by u-substitution.

5. $\int \sin^3 x \cos x \, dx$

6. $\int \sqrt{\cos x} \sin x \, dx$

7. $\int \tan^5(2x) \sec^2(2x) \, dx$

8. $\int \sin^7(2x) \cos(2x) \, dx$

9. $\int \tan\left(\frac{x}{2}\right) \sec^2\left(\frac{x}{2}\right) \, dx$

10. $\int \tan^2 x \sec^2 x \, dx$

Compute the following integrals using the guidelines for integrating powers of trigonometric functions. Use a CAS to check the solutions. (Note: Some of the problems may be done using techniques of integration learned previously.)

11. $\int \sin^3 x \, dx$

12. $\int \cos^3 x \, dx$

13. $\int \sin x \cos x \, dx$

14. $\int \cos^5 x \, dx$

15. $\int \sin^5 x \cos^2 x \, dx$

16. $\int \sin^3 x \cos^3 x \, dx$
17. $\int \sqrt{\sin x} \cos x \, dx$
18. $\int \sqrt{\sin x} \cos^3 x \, dx$
19. $\int \sec x \tan x \, dx$
20. $\int \tan(5x) \, dx$
21. $\int \tan^2 x \sec x \, dx$
22. $\int \tan x \sec^3 x \, dx$
23. $\int \sec^4 x \, dx$
24. $\int \cot x \, dx$
25. $\int \csc x \, dx$
26. $\int \frac{\tan^3 x}{\sqrt{\sec x}} \, dx$

For the following exercises, find a general formula for the integrals.

27. $\int \sin^2(ax) \cos(ax) \, dx$
28. $\int \sin(ax) \cos(ax) \, dx$.

Use the double-angle formulas to evaluate the following integrals.

29. $\int_0^\pi \sin^2 x \, dx$
30. $\int_0^\pi \sin^4 x \, dx$
31. $\int \cos^2 3x \, dx$
32. $\int \sin^2 x \cos^2 x \, dx$
33. $\int \sin^2 x \, dx + \int \cos^2 x \, dx$
34. $\int \sin^2 x \cos^2(2x) \, dx$

For the following exercises, evaluate the definite integrals. Express answers in exact form whenever possible.

35. $\int_0^{2\pi} \cos x \sin 2x \, dx$
36. $\int_0^\pi \sin 3x \sin 5x \, dx$
37. $\int_0^\pi \cos(99x) \sin(101x) \, dx$
38. $\int_{-\pi}^\pi \cos^2(3x) \, dx$
39. $\int_0^{2\pi} \sin x \sin(2x) \sin(3x) \, dx$
40. $\int_0^{4\pi} \cos(x/2) \sin(x/2) \, dx$
41. $\int_{\pi/6}^{\pi/3} \frac{\cos^3 x}{\sqrt{\sin x}} \, dx$ (Round this answer to three decimal places.)
42. $\int_{-\pi/3}^{\pi/3} \sqrt{\sec^2 x - 1} \, dx$

43. $\int_0^{\pi/2} \sqrt{1 - \cos(2x)} dx$
44. Find the area of the region bounded by the graphs of the equations $y = \sin x$, $y = \sin^3 x$, $x = 0$, and $x = \frac{\pi}{2}$.
45. Find the area of the region bounded by the graphs of the equations $y = \cos^2 x$, $y = \sin^2 x$, $x = -\frac{\pi}{4}$, and $x = \frac{\pi}{4}$.
46. A particle moves in a straight line with the velocity function $v(t) = \sin(\omega t) \cos^2(\omega t)$. Find its position function $x = f(t)$ if $f(0) = 0$.
47. Find the average value of the function $f(x) = \sin^2 x \cos^3 x$ over the interval $[-\pi, \pi]$.

For the following exercises, solve the differential equations.

48. $\frac{dy}{dx} = \sin^2 x$. The curve passes through point $(0, 0)$.
49. $\frac{dy}{d\theta} = \sin^4(\pi\theta)$
50. Find the length of the curve $y = \ln(\csc x)$, $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$.
51. Find the length of the curve $y = \ln(\sin x)$, $\frac{\pi}{3} \leq x \leq \frac{\pi}{2}$.
52. Find the volume generated by revolving the curve $y = \cos(3x)$ about the x -axis, $0 \leq x \leq \frac{\pi}{36}$.

For the following exercises, use this information: The inner product of two functions f and g over $[a, b]$ is defined by $f(x) \cdot g(x) = \langle f, g \rangle = \int_a^b f \cdot g dx$. Two distinct functions f and g are said to be orthogonal if $\langle f, g \rangle = 0$.

53. Show that $\{\sin(2x), \cos(3x)\}$ are orthogonal over the interval $[-\pi, \pi]$.
54. Evaluate $\int_{-\pi}^{\pi} \sin(mx) \cos(nx) dx$.
55. Integrate $y' = \sqrt{\tan x} \sec^4 x$.

For each pair of integrals, determine which one is more difficult to evaluate. Explain your reasoning.

56. $\int \sin^{456} x \cos x dx$ or $\int \sin^2 x \cos^2 x dx$
57. $\int \tan^{350} x \sec^2 x dx$ or $\int \tan^{350} x \sec x dx$